



The selection of pretreatment options for anaerobic digestion (AD): A case study in olive oil waste production

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HIGHLIGHTS

- Recalcitrant and polyphenols compounds in olive wastes make necessary their pretreatment.
- A screening on chemical and physical pretreatments was experimentally conducted.
- Biochemical Methane Potential (BMP) for each pretreatments were experimentally evaluated.
- The Energy Sustainability Index (ESI) parameter was used of screening the pretreatments.
- The best pretreatment using BMP and ESI parameters is the CaCO₃ addition.

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ABSTRACT

The aim of this work was to examine the effects of several pretreatment processes on anaerobic digestion (AD) by scoring the Biochemical Methane Potential (BMP) and the Energy Sustainability Index (ESI) of each process while taking into account the direct (heat and electricity) and indirect energy costs necessary to produce the chemicals that were used. The ESI is a measure of the ratio between the energy produced as methane during fermentation testing and that spent on pretreatment. The candidate procedures were applied to the main by-products of olive oil production: olive mill wastewaters (OMWW) and olive pomace (OP). Owing to their high organic concentration (COD > 250 g/L), high electrical conductivity, high concentration of polyphenols (0.5–5 g/L) and low pH (4–6), the biodegradation of these waste products is a difficult process with a high environmental impact. As OP and OMWW are composed of complex chemical substances such as cellulose and lignin, these products are hardly degraded pretreatment before AD becomes necessary. Based on BMP and ESI values, the most effective pretreatment was found to be the addition of CaCO₃, which resulted in a biogas production of 21.6 NL/L and an ESI = 14 (i.e., the energy obtained in the form of methane is 14 times that of the energy spent).

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1. Introduction

Olive oil is appreciated worldwide for its mild taste and certified beneficial proprieties for the human diet. Over 95% of the world's olive oil production occurs in the Mediterranean region. Producing approximately 4,650,000 pressed olives per year, Italy is the second leading olive oil producer in the world [1]. From an economic standpoint, olive oil is a highly profitable venture; however, it is expensive to manage the waste products that are generated during production, such as olive pomace (OP) and the olive mill wastewaters (OMWW). OP is the solid residue that remains after solvent extraction of the cold-pressed olives. Dry

OP is composed of lignin (37% w/w), cellulose and hemicelluloses (49.5% w/w), pulp-retained olive oil (7.5% w/w) and mineral components (6% w/w) [2,3]. OMWW is the aqueous effluent derived from olive oil production processes, including the washing operations of the drupes and plants. The typical composition of OMWW is 85–95% w/w water, 8.5–10% w/w organic compounds and 0.4–2.5% w/w mineral salts. The organic fraction contains 2–15% w/w phenol compounds, corresponding to a medium concentration of 0.5–5 g/l [4]. The above characteristics of OP and OMWW make these waste products a serious environmental threat to the Mediterranean region [5]. AD is reported to be the most promising technology for the degradation of OP and OMWW as it presents many appealing advantages including low levels of sludge generation, low or no nutrient requirements, reduction of greenhouse gas emissions, the production of a liquid fertilizer and

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the generation of renewable energy in the form of biogas containing methane [6].

Recently, several experimental tests were conducted to evaluate OP and OMWW biogas production yield. Operating in mesophilic conditions (35–38 °C) and in a batch mode, Tekin and Dalgic [7] and Agdag [8] obtained biogas production of 135 and 105 L/kg_{AVS} (variation of the VS content between the beginning and the end of the AD test) with methane concentrations of 75% v/v and 50% v/v, respectively.

Increased biogas production was obtained [2,1] when using reactors that were working in continuous mode with 300 and 251 L_{CH₄}/kg_{AVS}, respectively. Improving AD efficiency by employing pretreatment methods on waste products has been the focus of a large body of research in recent years. However, the applicability of different pretreatments to different substrates has not been assessed. In [9,10] suggestion that the lack of assessment is due to a defect of accepted standardized protocols for evaluating the efficiency of such pretreatments on AD was proposed, which is of high importance when energy production is considered [11]. In the case of the AD of olive oil production wastes, which are known to be very difficult to break down, some degree of pretreatment seems to be necessary. To this end, chemical compounds have been successfully used for the pre-treatment of ligno-cellulosic materials for the purpose of increasing the efficiency of the AD process. In [12] anaerobic batch testing on OP with the addition of 10 g/L Ca(OH)₂ and bentonite to bring the system up to pH = 6.5 are reported; the performance of the batch reactor at the end of the test was approximately 631 L_{CH₄}/kg_{AVS}. Azbar et al. [4] conducted different chemical pretreatments on the wastewaters generated from olive oil production by means of acid cracking followed by a coagulation–flocculation process using coagulation agents such as Al₂SO₄, FeSO₄ and FeCl₃. The cumulative results from these tests suggest that the pretreatment of waste products has good potential for improving the performance of AD. The application of aluminum salt was found to generate the highest gas production, indicating that OP degradation was significantly enhanced in the presence of this pretreatment. Follow up work [13] showed that organic matter is efficiently degraded via a Fenton-like reaction that uses FeCl₃ as a catalyst in the presence of hydrogen peroxide. OMWW contains phenolic compounds that are difficult to biodegrade, as they are originally synthesized by the olive plants as protection against pathogens or are formed during the olive oil extraction process and subsequently passed into the OMWW due to their high solubility [14]. The concentration of phenolic compounds in OMWW varies greatly, ranging from 0.002 to 80 g/L [15]. Though olive oil phenols have antioxidant, nutritional and anticancer properties [16], they are toxic to microorganisms and inhibit microbial growth via biostatic effects [17,18]. This suggests that a dephenolization process is necessary to enhance the performance of AD. Pekin et al. [19] showed that the pretreatment of OMWW using a process combining coagulation/flocculation and Fenton's oxidation is able to remove 80% of poly-phenols and 95% of the Total Solids (TS), respectively. In [20] are reported tests on four different adsorbent matrices (azolla, granular activated carbon, resins and zeolite) for their ability to dephenolize OMWW. The best removal of phenol compounds was obtained using granular activated carbon. Another study [21] compared acid cracking, lime precipitation, chemical coagulation, centrifugation and electrocoagulation processes on OMWW, taking into consideration both the biogas production and the economic costs of each of these methods. After balancing the COD, efficiency of phenol removal, and cost of operation, the centrifugation process was found to be the best choice.

The aim of this work was to build on these previous studies by experimentally verifying the effects of either chemical or physical pretreatment on AD fermentation. This was accomplished by testing the energy production, abatement of organic matter and lag

phase reduction in pretreated OP and OMWW samples to determine the most effective pretreatment option. As the purpose of AD is energy production, the operational energy costs associated with each pretreatment were also investigated. These costs were based on the consumption of direct energy (heat and electricity) during each process as well as on the indirect energy spent to produce the chemicals used. To this end, an Energy Sustainability Index (ESI) was employed to select the pretreatment method that was most efficient from an energy sustainability perspective.

2. Materials and methods

2.1. OP, OMWW and inoculum characteristics

OP and OMWW samples originated from Melendugno, a little town located near Lecce in the south of Italy. OP was derived from an olive production operation that employed a three-phase centrifugation process to separate the solid and aqueous phases of the oil. OMWW was obtained from the same production firm. Table 1 summarizes the main characteristics of the acquired OP and OMWW, which were experimentally evaluated at least three times. To guarantee that methanogenic bacteria were present in the reaction medium, bovine manure was used as inoculum as has been suggested in the literature [22]. Following evaluation of the relationship between bioreactor performance and microbial community structure during the treatment of actual wastewaters and biological wastes (such as municipal solid wastes, distillery wastewater, livestock manure and surplus sludge in a wastewater treatment plant) it has been reported that bovine manure is one of the best sources for methanogenic bacteria. The initial conditions of the inoculums were: pH = 7.3, density 888.4 kg/m³, and Total Solid (TS) and Volatile Solids (VS) concentrations of 221.4 g/L and 183.76 g/L, respectively.

2.2. Experimental apparatus

The experimental apparatus (Fig. 1) consisted of a 250 mL glass bottle, a 500 mL glass bottle and a plastic vessel connected by pipes. The 250 mL bottle contained a 100 mL mixture of OP and OMWW in an approximate 1:2 ratio and 10 mL (10% v/v) of bovine manure inoculum. The 1:2 ratio of OP to OMWW was chosen by taking into account the respective TS contents of these products (Table 1) so as to create a mixture containing approximately 10% w/w of TS. At the beginning of each experiment, the bottle was flushed with N₂ gas for 5 min to eliminate oxygen from the reaction medium and achieve anaerobic conditions. The 500 mL bottle served as the gasholder and was initially filled with water at pH = 2 to prevent dissolution of CO₂ (Fig. 1). As biogas is produced it moves from the first bottle into the second, replacing the water in the plastic vessel. The bottles serving as gasholders were exchanged when the water inside had almost completely moved into the plastic vessel due to biogas flow. The three bottles were agitated on a rotating shaker at 70 rpm in a thermostatic room kept at 30 ± 1 °C to avoid stressing the sensitive methanogenic microorganisms. A temperature of 30 °C was used in these experiments, as

Table 1
Chemical characteristics of OP and OMWW.

	OP	OMWW
Density (kg/m ³)	969.50 ± 41.2	989.40 ± 5.31
pH	6.75 ± 0.05	4.86 ± 0.01
TS content (g/l)	331.33 ± 6.81	12.04 ± 0.02
VS content (g/l)	305.60 ± 6.18	7.49 ± 0.21
Low Heat Value (kJ/kg)	25,504 ± 52	Negligible

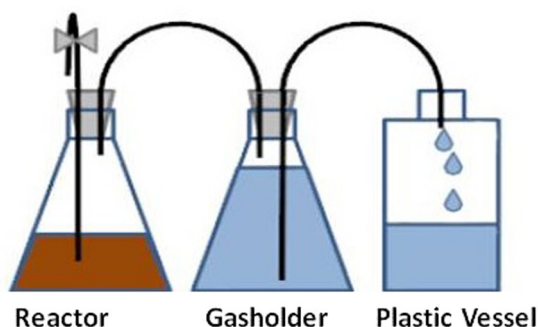


Fig. 1. The experimental apparatus.

this temperature is within the middle of mesophilic range. Furthermore, industrial work performed at higher temperatures leads to increased energy requirements in terms of heating the feed and thus to increased heat loss from the bioreactor wall due to convection heat transfer phenomena, which would decrease the net energy harvested by the AD process. All tests were conducted simultaneously in triplicate using three of the above described devices in parallel.

2.3. Analytical measurements

The total volume of biogas produced during fermentation testing was measured using the water-replacement method, as described above. This is one of the most commonly used laboratory methods for the quantification of small amounts of biogas as it produces low experimental uncertainty. The composition of gas produced in each test (H_2 , O_2 , CH_4 , CO_2 , CO and N_2) was determined in each gasholder by means of an off-line gas chromatographic analysis device (Varian, CP 4900) equipped with two columns: a Poroplot U column for CO_2 determination (85 °C injection temperature, 200 kPa, Argon as the carrier) and a Molecular Sieve type for H_2 , CH_4 , CO , O_2 and N_2 determinations (95 °C injection temperature, 200 kPa, Helium as the carrier). The pH of the system was determined via a pH meter (CRISON model micropH 2001), and TS, VS and density were determined according to the Standard method [23]. The Low Heat Value (LHV) of the OP and OMWW were experimentally determined using the bomb calorimetric PARR 1261 (Parr-Instrument USA). Measurements of LHV were repeated in triplicate to achieve statistical significance and to appropriately evaluate the uncertainty of the experimental results.

2.4. Biochemical Methane Potential (BMP) experimental procedures

Several batch digestion tests were performed under mesophilic conditions, as reported in Table 2. Each test was conducted in triplicate simultaneously. All tests used a feeding mixture comprised of OP and OMWW with a TS content of approximately 10% w/w (see Table 3 for exact values). The mixture was obtained by adding 33 g of OP to 77 g of OMWW. A concentration of approximately 120 g/l was also used in cases when OP was diluted with water. This dilution was also considered a pretreatment process (Table 3). Following pretreatment, the broth was divided into three aliquots and each was inoculated with bovine manure. The initial pH for all the tests was adjusted to approximately 7 using 2 N NaOH or 2 N HCl when necessary. During the tests the pH was not controlled and was therefore free to move according to the bioreaction. The tests were stopped when biogas production shut down (approximately 60 days). The gas composition was determined when each gasholder was exhausted. To obtain a base comparison for the performance of the pretreated mixtures, a test using OP and OMWW with no pretreatment was carried out under the same experimental conditions.

3. Pretreatment tests

Several pretreatment processes were evaluated (Table 2): basic pretreatment (BP); salt pretreatment using $FeSO_4$ (SP_{FeII}), $FeCl_3$ (SP_{FeIII}), $MnSO_4$ (SP_{Mn}) and $CaCO_3$ (SP_{Ca}); physical pretreatment using ultrasound (USP); and double combinations of the aforementioned pretreatments, such as ultrasound plus $FeCl_3$ (CP_{FeIII}) or ultrasound plus $CaCO_3$ (CP_{Ca}). The energy produced as methane during each test was compared to that obtained with the no pretreatment mixture, NP (Table 2). All pretreatment tests were conducted in a thermostatic chamber at 30 ± 1 °C, the methodology of which is briefly described in the following sections. The main characteristics and abbreviations for each pretreatment process are summarized in Table 2.

3.1. Basic pretreatment (BP)

BP tests were conducted by adding 2 N NaOH to the mixture until a pH of 12 was reached, then stirring the mixture at 70 rpm for 24 h at 30 ± 1 °C. The broth was neutralized with 2 N HCl before the addition of inoculum. Amidst the repertoire of available chemical pretreatments [24], this basic pretreatment is recognized for its efficacy in improving hydrolysis efficiency, which is the first step in the AD [25] of materials that contain a large percentage of lignin and cellulose, such as OP. Fan et al. [26] suggested that this treatment removes lignins primarily via a saponification

Table 2
Descriptions and abbreviations of the pretreatments tested.

Pretreatments	Abbreviations
No pretreatment OP + OMWW	NP
Dilution with water	WP
Basic pretreatment	BP
Salts pretreatments	
$FeSO_4$	SP_{FeII}
$FeCl_3$	SP_{FeIII}
$MnSO_4$	SP_{Mn}
$CaCO_3$	SP_{Ca}
Ultrasonic pretreatment (8000 kJ/L)	USP^a
Ultrasonic pretreatment (750 kJ/L)	USP^b
Ultrasonic and $FeCl_3$ pretreatment	CP_{FeIII}
Ultrasonic and $CaCO_3$ pretreatment	CP_{Ca}

Sonication dose: ^a8000 kJ/L; ^b750 kJ/L.

Table 3
Experimental results.

Test	pH _{in}	pH _{fin}	TS reduction (%)	VS reduction (%)	Biogas (NL/L)	CH ₄ (% v/v)
NP	7.1 ± 0.2	5.90 ± 0.03	35.44 ± 1.64	34.84 ± 0.52	0.36 ± 0.06	51.60 ± 4.02
WP	7.1 ± 0.3	7.01 ± 0.01	43.06 ± 0.65	43.87 ± 0.26	13.56 ± 0.18	74.31 ± 1.96
BP	7.1 ± 0.4	7.56 ± 0.02	47.87 ± 0.92	55.66 ± 0.52	18.98 ± 0.91	86.3 ± 2.31
SP _{Fell}	7.1 ± 0.5	7.02 ± 0.02	40.36 ± 0.54	49.04 ± 0.47	10.74 ± 1.31	81.88 ± 3.22
SP _{Mn}	7.1 ± 0.7	5.77 ± 0.02	17.35 ± 1.01	23.56 ± 1.20	0.27 ± 0.05	40.00 ± 2.65
SP _{FellII}	7.1 ± 0.6	6.98 ± 0.03	57.73 ± 0.74	65.07 ± 1.13	19.89 ± 3.53	84.25 ± 2.75
SP _{Ca}	7.1 ± 0.8	7.18 ± 0.04	31.09 ± 0.24	31.18 ± 0.91	21.57 ± 3.10	83.02 ± 2.36
USP ^{a)}	7.1 ± 0.9	7.30 ± 0.02	34.45 ± 0.17	38.68 ± 0.32	22.17 ± 1.18	84.93 ± 1.92
USP ^{b)}	7.1 ± 0.9	7.10 ± 0.01	36.01 ± 0.58	33.10 ± 0.73	5.10 ± 1.00	55.2 ± 2.00
CP _{Ca}	7.1 ± 0.10	7.32 ± 0.02	41.38 ± 1.08	52.11 ± 0.67	16.48 ± 0.53	81.00 ± 4.01
CP _{FellII}	7.1 ± 0.11	7.11 ± 0.02	36.03 ± 0.29	33.69 ± 0.19	8.75 ± 0.67	82.37 ± 2.84

Sonication dose: ^{a)}8000 kJ/L; ^{b)}750 kJ/L.**Table 4**
Yield of methane production with uncertainty.

Test	Y (L _{CH₄} /kg _{ΔVS})	Y (L _{CH₄} /kg _{VS0})
NP	6.67 ± 4.04 (60%)	1.82 ± 0.53 (29%)
WP	258.33 ± 79.80 (31%)	103.67 ± 9.17 (9%)
BP	346.97 ± 117.30 (34%)	173.32 ± 24.16 (14%)
SP _{Fell}	190.84 ± 71.14 (37%)	86.23 ± 19.59 (23%)
SP _{Mn}	5.79 ± 3.59 (62%)	1.04 ± 0.31 (30%)
SP _{FellII}	282.01 ± 91.48 (32%)	176.81 ± 47.03 (27%)
SP _{Ca}	748.22 ± 381.94 (51%)	193.38 ± 45.86 (24%)
USP ^{a)}	537.68 ± 222.67 (41%)	179.36 ± 22.10 (12%)
USP ^{b)}	101.19 ± 51.78 (51%)	28.11 ± 7.95 (28%)
CP _{Ca}	473.59 ± 224.93 (47%)	196.49 ± 35.79 (18%)
CP _{FellII}	212.42 ± 53.98 (25%)	75.77 ± 11.54 (15%)

The percentage values in the parenthesis indicate the relative uncertainty.

Sonication dose: ^{a)}8000 kJ/L; ^{b)}750 kJ/L.

reaction of the intermolecular ester bonds that crosslink xylon hemicelluloses and other components. The mechanism of action behind this phenomenon appears to be caused by an increased porosity of the ligno-cellulosic materials. Generally, NaOH treatment appears to create several beneficial effects towards the bio-degradation of ligno-cellulosic materials: an increased internal surface area, a decreased degree of polymerization and crystallinity, and a destruction of the structural links between the carbohydrates and lignin [26].

3.2. Salt pretreatments (SP)

OP and OMWW mixtures were treated with FeSO₄ (SP_{Fell}), MnSO₄·H₂O (SP_{Mn}), FeCl₃ (SP_{FellII}) or CaCO₃ (SP_{Ca}). Suitable quantities of solid salt were added to achieve a 5 g/L salt concentration within the mixture. The reaction mediums were treated with 2 N HCl until a pH of 2 was reached and then mixed at ambient temperature for 30 min at 70 rpm. Following this, the specific salt being tested was added at the desired concentration. The mixtures were then held at 70 rpm and 30 ± 1 °C for 24 h. Finally, 2 N NaOH was added until a neutral pH was reached. Acid was not used to adjust the pH in the case of CaCO₃ pretreatment; instead, a pH of approximately 7 was reached by using 2 N NaOH after the addition of the salt. The decision to use a salt concentration of 5 g/L was made based on previous studies that demonstrate it as being the most effective for optimizing biogas production [12,4,27].

3.3. Ultrasonic pretreatment (USP)

Sonication is a well-known method of breaking cellular walls and has recently been utilized as a pretreatment method to increase the production of biogas during AD [28]. Ultrasonic pretreatments were carried out in an ultrasonic bath (Ultrasonic Cleaner model CP823) under the following conditions: 30 °C,

atmospheric pressure and the application of 1.8 kW of power for varying lengths of time, such that different doses of ultrasonic rays could be applied (Table 2). Using a glass container, the mixture being tested was inserted into the ultrasonic basin, which contained tap water. Our aim was to evaluate whether the ultrasonic pretreatment could break down large molecules into smaller pieces that could be more easily digested by microorganisms. Two tests were performed to evaluate the dose to be applied in terms of energy per unit of volume of fermenting broth (kJ/L). The efficacy of ultrasound becomes saturable at high doses [29] depending on the concentration of TS in the fermenting broth. Therefore, an initial test was performed using a very high dose of 8000 kJ/L on a mixture of OP and OMWW. This tested was used to evaluate the *BMP* (see Section 2.4) of the pretreated mixture so as to determine the maximum effect that could be obtained for methane production. A second ultrasonic pretreatment was carried out at a dose based on the energy produced in the first *BMP* test: applying the energy sustainable criterion (i.e., *ESI* = 1; Section 5), a value of 750 kJ/L was chosen for evaluation. The purpose of this test was to compare the *BMP* value with that generated during the first test (Table 3). In this manner one experimentally acquires the highest dose needed to produce maximum energy and the lowest dose needed to maintain energetic sustainability (Table 3).

3.4. Combined Pretreatments (CP)

Two different pretreatment combinations, ultrasonic plus CaCO₃ pretreatment (CP_{Ca}) and ultrasonic plus FeCl₃ pretreatment (CP_{FellII}), were tested as a means of increasing biogas production. For these tests the medium was first treated with ultrasonic rays at the highest dose of 8000 kJ/L and then with the salts, as described above.

4. Data analysis

4.1. Cumulative gas production parameters

To evaluate *BMP* parameters, a regression curve was calculated to describe the biogas produced in our experimental set up. The Gompertz model was employed [30], as its sigmoidal function is one of the most useful for characterizing saturation kinetics. During saturation, biogas production proceeds slowly at early time points and continues for a long time, until the end of the reaction is reached (producing an asymptotic value), it is possible to describe exponential growth using the following Gompertz equation:

$$G = G_{\max} \exp \left\{ - \exp \left[\frac{R_{\max} * e}{G_{\max}} (\lambda - t) + 1 \right] \right\} \quad (1)$$

where G (L/L) represents the cumulative volume of biogas produced during time t , R_{max} (L/L day) represents the maximum value of the biogas production rate, G_{max} (L/L) represents the total quantity of the biogas produced as an asymptotic value and λ (day) represents the duration of the lag phase. The cumulative experimental data produced during each test were described with Eq. (1). This allowed us to estimate the three BMP parameters of each test: G_{max} , R_{max} and λ . The parameters were estimated by employing a non-linear best-fit procedure to the mean values of the three repeated tests (Systas Software Inc. Sigma Plot) [31]. Additionally, by combining G_{max} values and Eq. (1), the Half Period Time (HPT) was calculated for each test, which represents the time at which a quantity of biogas equal to the $G_{max}/2$ was produced i.e., the time necessary to reach 50% of the yield and indicates the time at which the production rate is equal to its maximum value, R_{max} . This parameter was evaluated as different pretreatments can produce equivalent yields while exhibiting different kinetics (e.g., Fig. 2b).

4.2. Energy conversion parameters

To evaluate the efficacy of pretreatment on OP and OMWW residues, the energy produced during each pretreatment and BMP evaluation test, was compared with that generated from samples that were not subjected to pretreatment. Additional information on this approach can be found in [34]. Two energy parameters were used to score the pretreatments. The first parameter evaluated was the energy conversion efficiency (ξ), which evaluates

the ratio of energy produced as methane during pretreatment/bio-reaction to the available amount of energy contained within the substrate being evaluated. ξ is defined as:

$$\xi = E_p / E_0 * 100 \quad (2)$$

where E_p represents the total energy produced as methane (kJ/L) and E_0 represents the initial energy contained within the substrate, considering the concentration of OP plus OMWW present in the broth (kJ/L). The produced energy (E_p) was calculated using:

$$E_p = G'_{max} / 22.4 * LHV_{CH_4} \quad (3)$$

where G'_{max} represents the total volume of methane produced at normal conditions (1 atm and 0 °C) (NL/L) and is based on the mean concentration of methane in biogas, 22.4 is the equivalent volume of mol (L/mol), and LHV_{CH_4} represents the Low Heat Value of the methane per mol unit (800.32 kJ/mol). When calculating E_0 , it is necessary to consider only the waste product energy that can be utilized by microorganisms for cellular activity [33], which is a difficult task. As the purpose of this calculation is to score the outcomes of different pretreatment processes performed on the same substrates, a good approximation of E_0 could be reached using the LHV of the substrates. Eq. (4) evaluates E_0 in this manner:

$$E_0 = LHV_{OP} * TS_0 * 10^{-3} \quad (4)$$

where LHV_{OP} (kJ/kg) represents the Low Heat Value of OP as reported in Table 1 and TS_0 is the initial concentration present in the reaction medium in g/L. It is important to note that not all of

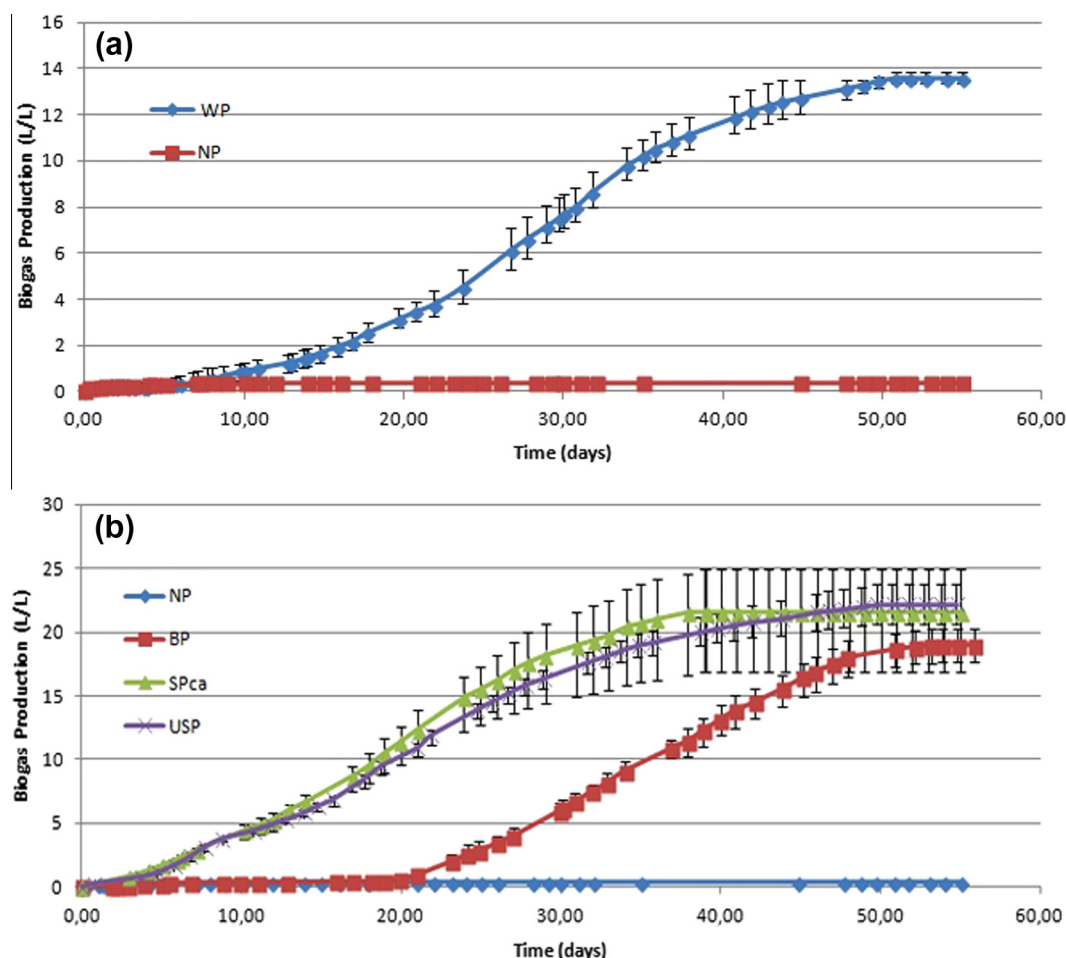


Fig. 2. Cumulative biogas production for the most effective tested pretreatments. (a) Biogas production comparison of NP and WP tests; (b) biogas production comparison of the most effective pretreatments (BP, SPCa, USP) and NP test.

the energy contained within the substrates is available for methane production. Indeed, not all C–C or C–H bonds are of the same type and microorganisms can only easily degrade the bonds of simple molecules i.e. the edible one. Furthermore, only a portion of this energy can be harvested as methane using AD technology, as a fraction of energy is consumed during biological metabolism, a fraction is released as heat, a fraction is stored within newly divided cells and a fraction is embedded within the metabolites produced during the reaction. If glucose was instead used as the source of C, as is generally accepted by biologists, then all of the chemical energy would be considered consumable. The available energy in glucose is 2880 kJ/mol, which is its LHV value. Considering that a combination of consumable and non-consumable energy is present within any substrate, thus experimentally it is possible to measure only the total energy as LHV. Unfortunately, current technology cannot experimentally measure the amount of bioconsumable energy present in a substrate. The aim of this work was to evaluate the efficiency of AD technology and whether pretreatment of waste products can increase the production of methane gas. Eq. (2) is able to give global and effective information on AD and other technologies by evaluating the energy produced as methane (*usable energy*) in relation to the energy contained within the waste products (*available energy*) as LHV.

The second parameter evaluated was the efficacy (η), which relates the efficiency of testing performed following pretreatment to that performed without pretreatment. η was calculated as:

$$\eta = \xi_P / \xi_{NP} \quad (5)$$

where ξ_P represents the efficiency obtained using pretreated broth and ξ_{NP} represents the efficiency obtained using broth without pretreatment. Thus, η is a ratio of energy obtained following pretreatment to energy obtained without it. This allows for easy evaluation of the effectiveness of a given pretreatment: η values less than 1 indicate that pretreatment had a negative effect on efficiency (perhaps through the production of an inhibitory substance); η values approximating 1 indicate that a given pretreatment had no effect on the bioreaction; and η values greater than 1 indicate that pretreatment increased the quantity of energy produced as methane. In this way, η permits comparative scoring of different pretreatment options.

4.3. Energy Sustainability Index

As AD is a viable candidate technology for the production of societally “useful energy” it is important to verify its energetically sustainability, such as by comparing the energy produced by the system in the form of biogas with the energy spent during the process. Some analysis on the energetic sustainability of AD has been conducted in the past [32]. In this work we evaluated which

waste product pretreatments could be most effectively coupled with AD to select the most energetically effective method of processing OP and OMWW by comparing *BMP* values for the energy spent during pretreatment to that obtained during fermentation, to evaluating the efficacy of each system. To achieve this aim, the Energy Sustainability Index (*ESI*) was employed as:

$$ESI = E_{ob} / E_{tot} \quad (6)$$

in which E_{ob} represents the energy obtained as methane during fermentation, and E_{tot} represents the total energy spent for the pretreatment. $ESI > 1$ indicates the possibility of an energetically sustainable process that merits a more detailed energetic sustainability analysis [32]. Alternatively, $ESI < 1$ indicates a process that is not sustainable, even if it increases energy production (i.e., $\eta > 1$). For each pretreatment considered here, the direct energy spent to conduct the test (e.g., heat and electricity) was considered in addition to the energy spent to produce the chemicals used:

$$E_{tot} = E_h + E_{EE} + E_{Ch} \quad (7)$$

where E_h represents the energy spent to heat the broth to the working temperature of 30 °C:

$$E_h = \rho C_p (T_a - T_w) \quad (8)$$

An ambient temperature of $T_a = 10$ °C was considered to be a representative average of summer and winter conditions, and $T_w = 30$ °C was considered the working temperature. ρ represents the mass present per unit of volume of broth and C_p represents the specific heat value. Assuming that the reaction medium is similar to water, we obtain $C_p = 1$ (kJ/kg °C) and $\rho = 1$ (kg/L). E_{EE} represents the electrical energy spent and is obtained by multiplying the power of the ultrasonic bath by the duration of time used for each test. E_{Ch} represents the energy spent to produce the chemicals (NaOH, HCl and salts) used. These specific energies were evaluated as Global Energy Requirements (*GER*; kJ/kg) using data from the Ecoinvent (2010) database. The values reported in Table 5 were obtained by multiplying the specific *GER* by the quantity of chemical used for each test.

5. Results and discussion

5.1. Dilution with water and OMWW

In Fig. 2 the experimental results of biogas production during the NP test are reported (L/L) in addition to the results obtained following dilution of OP + OMWW with water (WP). In the NP test the OP was diluted with OMWW. As shown (Fig. 2a), minimal biogas was produced during NP and production was concentrated over the first 10 days. The percentage of methane in the biogas was approximately 50% v/v, which is considered to be low. The

Table 5
Evaluation of energy spent for chemicals in the pretreatment.

	HCl (g)	NaOH (g)	CaCO ₃ (g)	FeCl ₃ (g)	MnSO ₄ (g)	FeSO ₄ (g)	Ech (kJ)
NP							
WP							
BP	6.05	0.55					114.69
SP _{FeII}	6.88	0.55				0.55	131.99
SP _{Mn}	6.88	0.55			0.55		133.37
SP _{FeIII}	6.88	0.55		0.55			138.86
SP _{Ca}			0.55				0.42
USP ^{a)}							
USP ^{b)}							
CP _{Ca}			0.55				0.42
CP _{FeIII}	6.88			0.55			138.85
GER (kJ _{eq} /kg)	18,413	5,980	768.03	16,322	6,345	3,836	

Sonication dose: ^{a)}8000 kJ/L; ^{b)}750 kJ/L.

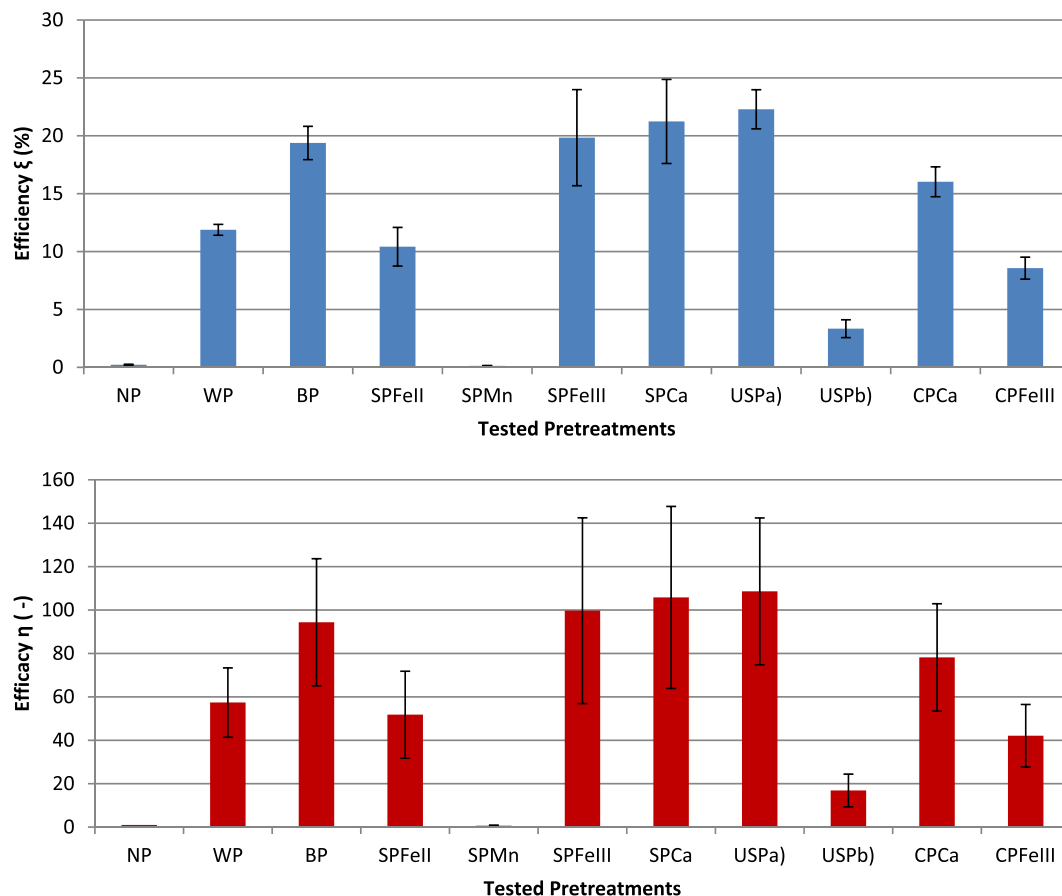


Fig. 3. Efficiency and efficacy results for each pretreatment tested (including uncertainty).

values of TS, VS reductions and ξ were 35.44%, 34.84% (Table 3) and 0.22% (Fig. 3), respectively, indicating that at the end of the test the reaction mixture was still rich in organic matter available for conversion into methane. It is likely that the high poly-phenol content of the OMWW inhibited the methanogenic bacteria activity [16,17] and that the lignocelluloses nature of the largest organic structures contributed to the low biodegradability of OP and OMWW. Additionally, the final pH value of 5.9 (Table 3) is a clear indication that the production of methane was a difficult process under these conditions. A pH drop is often caused by an increase in volatile fatty acids (VFA) during the first phase of AD, when hydrogen-producing bacteria have faster reaction kinetics than methanogenic bacteria. Consequently, the complex substances in the reaction medium are reduced to simpler molecules, mainly short chains of organic acids (VFA), which act as the substrates for methanogenic bacteria during subsequent reaction phases. If the un-dissociated VFA concentration becomes greater than 50 mg/l, methanogenic bacteria become inhibited and methane production drops [35]. Low methanation activity will also slow the progress of AD.

The dilution of olive oil waste residues with tap water (WP) was able to increase both biogas production (Fig. 2a) and CH_4 concentration (Table 3). The biogas produced was approximately 13.5 L/L with a methane content of 74% v/v. The addition of water to the feed mixture mitigates the acidity of the feed and likely decreases the concentration of polyphenols within the reaction medium. As reported in Table 6, the lag phase (λ) is high, almost 2 weeks, and HPT is almost 1 month. These facts demonstrate that without employing pretreatment and only diluting the waste products with tap water the kinetics of the bioreaction is slow, likely because of the presence of complex molecular structures within

Table 6

Estimated Gompertz parameters.

Test	$G'_{\max}$ (NL/L)	$R_{\max}$ (NL/L d)	λ (days)	HPT (days)	R^2
NP	0.36 ± 0.06	8.75	n.a.	1.26	0.82
WP	13.55 ± 0.18	73.86	12.7	28.14	0.95
BP	19.01 ± 0.91	104.82	22.14	34.67	0.94
SP _{FeII}	10.74 ± 1.31	92.50	20.42	35.44	0.91
SP _{Mn}	0.27 ± 0.05	6.52	n.a.	1.26	0.94
SP _{FeIII}	19.85 ± 3.53	127.19	17.06	32.76	0.95
SP _{Ca}	21.60 ± 3.10	120.53	6.05	19.38	0.94
USP ^{a)}	22.23 ± 1.18	103.12	5.6	20.87	0.97
USP ^{b)}	5.10 ± 1.00	21.74	6.2	17.32	0.95
CP _{Ca}	16.75 ± 0.53	93.23	8.4	18.95	0.93
CP _{FeIII}	8.80 ± 0.67	40.11	1.3	80.88	0.95

Sonication dose: ^{a)}8000 kJ/L; ^{b)}750 kJ/L.

n.a. stand for "not available".

the reaction environment that inhibit the microorganism activity. Intriguingly, the WP test produced the second highest *ESI* value (*ESI* 8.62; Table 7), though this was due to the low energetic requirements of the pretreatment, as it consists only in the addition of water. In fact the only energy consumption required during this pretreatment is producing enough heat to reach the working temperature. Thus, dilution of waste products with water is one of the most competitive pretreatment options among those tested in the present study.

5.2. Basic pretreatment

In Fig. 2b the cumulative biogas production curves for BP, SP_{Ca} and USP are reported along with the results of the NP test. The

Table 7
Energetic Sustainability Index (ESI) values of the tested pretreatments.

Test	E_h	E_{EE}	E_{ch}	E_{tot}	E_{ob}	ESI
NP	5.02			5.02	0.80	0.16
WP	5.02			5.02	43.27	8.62
BP	5.02		114.69	119.71	70.58	0.59
SP _{Fell}	5.02		131.99	137.01	37.96	0.28
SP _{Mn}	5.02		133.37	138.39	0.47	$3 \cdot 10^{-3}$
SP _{FellII}	5.02		138.86	143.88	72.27	0.50
SP _{Ca}	5.02		0.42	5.44	77.35	14.21
USP ^{a)}	5.02	891		896.02	81.21	0.09
USP ^{b)}	5.02	82		7.02	12.18	0.14
CP _{Ca}	5.02	891	0.42	896.44	58.38	0.07
CP _{Fell}	5.02	891	138.85	1,034.87	31.22	0.03

All the values are expressed in kJ considering: the TS, the volume, the quantities of chemicals and the electrical energy spent for each test.
Sonication dose: ^{a)}8000 kJ/L; ^{b)}750 kJ/L.

BP test resulted in robust production of biogas (19 L/L) with high methane content (86.3% v/v; Table 3). These results can be explained by the fact that the two most abundant chemical components in OP are cellulose, a biopolymer that consists of thousands of glucose units, and lignin, an insoluble polymer of high molecular weight that is composed of complex carbohydrate chains. The alkaline hydrolysis mechanism that occurs as a consequence of NaOH treatment is believed to involve a saponification reaction of inter-molecular ester bonds [24,27]. Pretreatment of lignocellulosic materials with dilute NaOH leads to swelling and provokes different effects: an increase in the internal surface area; a decrease in the degree of polymerisation and crystallinity; separation of the structural links between lignin and carbohydrates; and the disruption of lignin structure. As a consequence, the basic pretreatment exploits the lignocelluloses structures present in the reaction mixture, as evidenced by the efficiency value of BP test ($\xi = 19.38$; Fig. 3). The negative aspects of the BP test are demonstrated in the high values produced for λ and HPT (22.14 and 34.67 days, respectively; Table 6). Thus we can conclude that although BP increases energy production, the reaction kinetics remain slow. It is also noteworthy that BP is not competitive from energetic point of view, as the ESI (0.59; Table 7) for this process demonstrates that it is negatively affected by the high energy consumption required to produce NaOH, as well as by that spent on HCl during the pretreatment (Table 5).

5.3. Pretreatment using salt

When evaluating the quantity of methane energy obtained following pretreatments employing the use of various salts, it is noteworthy that despite the exception of the SP_{Mn} test, high values were obtained (Table 4). The SP_{Mn} test performed poorly with regards to biogas production (0.27 L/L) and methane content (40% v/v; Table 3). As a consequence, TS and VS reductions were 17.35% and 23.56%, respectively (Table 3). The low efficiency of the process ($\xi = 0.13$) was also inferior to that obtained with the NP test (Fig. 3). Considering the final pH value obtained after performing this test (5.77; Table 3), the poor efficiency may be due to the system going into an acidogenic phase, which would strongly inhibit the activity of methanogenic bacteria. This phenomenon was likely due to the presence of sulfate ions derived from the salt MnSO₄, which was added during the pretreatment process. As demonstrated in [35], sulfate is reduced preferentially to H₂S, resulting in a decreased pH within the medium. The same phenomenon might also be explained by thermodynamics: sulfate degrading microorganisms are dominant in the presence of SO₄²⁻ as they generate higher free energy and do not require a symbiotic partner. Indeed, sulfate degradation requires a free energy of -154 kJ mol^{-1} , while the methanation of carbon dioxide and

hydrogen requires $-135.4 \text{ kJ mol}^{-1}$ and the methanation of acetic acid $-30.9 \text{ kJ mol}^{-1}$ [36]. The poor performance of the SP_{Mn} test did not permit evaluation of λ and HPT as biogas production never really started.

SP_{Fell} and SP_{FellII} testing involves a pre-acid cracking followed by the respective addition of two poly-electrolyte salts, FeSO₄ and FeCl₃. Previous research [37,38] has demonstrated the beneficial effect of using cationic polyelectrolytes to promote the Fenton reaction of organic matter. This process increases the reduction of VS and TS, as reported in Table 3, as well as the removal of the majority of poly-phenolic compounds (which are colloidal substances). Polyelectrolyte salts are metal cations and thus have a destabilizing action by way of their positive charge, which neutralizes the negative charge of colloids. In particular, the SP_{FellII} test demonstrated robust production of biogas (approximately 20 NL/L) with a methane content of 84.25% v/v (Table 3). The value $\xi = 20$ was one of the best results obtained in this study and suggests that FeCl₃ pretreatment improved the energy production of the system by approximately 100 fold. This can be further demonstrated by the value $\eta = 99.72$ (Fig. 3). The high energy required to produce the chemicals used in SP_{FellII} (138.86 kJ; Table 5) results in an ESI value of less than 1 (ESI = 0.50; Table 7). This suggests that the energy obtained as methane during this test is not sufficient to cover the energy spent on FeCl₃ pretreatment despite its high efficacy.

The SP_{Ca} test performed similarly to the SP_{FellII} test. As reported in Table 3, the biogas production was high (21.57 N L/L) and had good methane content (83.02% v/v). The ξ and η values (21.24 and 105.81, respectively) obtained during SP_{Ca} testing (Fig. 3) were the best among the tests conducted during this study. An important effect observed during SP_{Ca} testing was the reduction of the lag phase λ (Table 6), which decreased from approximately 13 (as occurs following simple dilution with water) to 6.1 days, indicating that robust bioreaction kinetics were achieved. A final positive aspect of SP_{Ca} testing was the ESI obtained during this process (14.21; Table 7), which was the highest ESI value obtained. This favorable ESI value was likely obtained because the addition of NaOH was not required to control the pH during this reaction and because the production of CaCO₃ does not require many energy (Table 5).

The robust performance of the CP_{Ca} test can be explained by taking into account that at a calcium carbonate concentration of approximately 5 g Ca²⁺/L there is an increase in both the formation of biofilm on lignocelluloses materials and in biomass growth [35,39]. Additionally, calcium is known to be essential to the growth of certain strains of methanogens and for the formation of microbial aggregates. In [39] is reported that Ca²⁺ concentrations of up to 7 g/L have no inhibitory effects on anaerobic digestion. Furthermore, we should not disregard that the presence of CaCO₃ results in a buffering action that can prevent the acidification of the reaction medium in response to VFA production during the acidogenic phase. For the SP tests evaluated, λ and HPT remained high. The only exception to this was the SP_{Ca} test where the values of λ and HPT were 6.1 and 19.38 days, respectively (Table 6). These results confirm that calcium positively influences methanogen growth kinetics.

5.4. Ultrasonic pretreatment

The literature reports conflicting results on the effects of ultrasonic rays on AD. Liu et al. [40] assert that ultrasonic pretreatment does not result in improved methane production during the anaerobic digestion of organic substrates, while Lee et al. [28] show a beneficial effect on biogas production when using filamentous algal as substrate. The experimental results generated in this work show that pretreatment using a high dose USP^{a)} (8000 kJ/L)

Table 8

Methane yields generated in our experiments as compared to those found in the literature.

References	Test description	Methane yield (L_{CH_4}/kg_{AVS})	Methane yield (L_{CH_4}/kg_{VSO})
[8]	OP codigestion	160	58
[15]	OP + OMWW without pretreatment	17	1.4
[8]	OP + OMWW with $AlSO_4$	36	12
[12]	OP + bentonite	800	241
[5]	Two phase OP	480	27
[3]	OP + water AD	294	84
[7]	OP + water	240	76
This work	NP	7	2
This work	WP	258	104
This work	SP _{Ca}	748	193

produced the best values for both ξ and η (22.29 and 108.62, respectively; Fig. 3). This treatment also led to extremely robust biogas production (22.17 NL/L) with good methane content (84%, 93%). The ultrasonic pretreatment likely contributes to the destruction of complex chemical structures (lignin, cellulose) within OP, which can increase the surface area available for bacterial degradation. This would enhance the mass transfer phenomena which are always present in a heterogenic reaction system, as demonstrated by the decreased kinetic value of $\lambda = 5.6$ days (Table 6). When pretreatment was performed using a dose of 750 kJ/L USP^b, biogas production was increased by approximately 14-fold over NP testing (Table 3). This confirms the beneficial effects of the sonication process on biogas production. However, USP requires a very high energy cost in terms of electricity (Table 7) and thus pretreatment using sonication is not beneficial from an energetic point of view (having *ESI* values of 0.09 and 0.14 for USP^a) and USP^b, respectively). One possibility for achieving energetic sustainability in such a system would be to use a reduced dose (e.g., 3000 kJ/L). However, to obtain an *ESI* = 1 in such a case it would be necessary to produce 330 kJ of methane energy from 110 mL of mixture. This value is 4-fold higher than that obtained by treating the mixture with a high dose (8000 kJ/L; Table 7). We conclude that ultrasonic pretreatment is beneficial towards energy production by AD but is unsustainable from an energetic point of view as the energy recovered from the system is less than that which is expended.

5.5. Combining pretreatments

The respective combining of sonication tests CP_{Ca} and CP_{FeIII} with CaCO₃ and FeCl₃ pretreatment unexpectedly resulted in poorer performance than when performing the tests individually. In fact, CP_{Ca} produced values of $\xi = 16$ and $\eta = 78.18$ compared to $\xi = 21$ and $\eta = 105$ in the case of SP_{Ca} (Fig. 3). Similarly, the performance of CP_{FeIII} was made worse compared to either USP^a) or SP_{FeIII} testing alone. As the *ESI* parameters generated from the combined tests produced values measuring less than 1 (Table 7), the combined tests are considered energetically unsustainable.

Finally, to compare the present results with those in the literature, Table 4 shows the evaluation of the specific yields for each test reported alongside its absolute and relative uncertainty. The relative uncertainty was high for each test (mean value 50%). This may be due changes within the complexity of system (e.g., different chemicals residues after different pretreatment and different microorganisms population balance growth) when evaluated at different experimental sites; in addition the necessary mathematical manipulation of the experimental data also introduces uncertainty in the results. The experimental uncertainty for the present results was calculated based on the dispersion of three repeated tests. In our opinion it is of fundamental importance to

evaluate the uncertainty when comparing different results, as well as for scale-up purposes. Table 8 shows the comparison of the result of this study with values taken from the literature. Unfortunately, the uncertainty was not possible to calculate in the referenced paper hence the comparison was made only by mean values. The best results were obtained from pretreating OP and OMWW with CaCO₃ is in agreement with the referenced work, particularly when examining of the it ranges in the upper part. It is also noteworthy that the values produced for the pretreatment testing that involved only the dilution of the waste products with water agree with those found in the literature.

6. Conclusion

OP and OMWW are wastes rich in organic matter and contain complex and inhibitory substances that can slow down or prevent anaerobic fermentation. For this reason, pretreatment processing of these compounds is essential. In this work, a variety of pretreatments were evaluated. We demonstrated that under the right conditions olive oil waste can be degraded anaerobically into biogases with high methane contents. Even the simple dilution of OP with tap water (WP) led to beneficial improvements in anaerobic digestion, improving the energy produced by over 57-fold compared to untreated waste (NP). However, for the purpose of methane production, the most effective pretreatments were found to be the addition of either CaCO₃ or FeCl₃, as well as the ultrasonic pretreatment. In fact, the ultrasonic pretreatment had an efficacy of 108, which was the highest among our panel of pretreatment processes. Amongst the three most effective pretreatments, CaCO₃ addition was the only method capable of producing a positive energy balance between energy obtained and energy spent, likely because the production of CaCO₃ is a low energy process in itself. This was evaluated by the *ESI* parameter, i.e., the ratio between the energy produced and that spent to conduct the test. The effectiveness of carbonate addition is also evident when comparing the methane yield obtained in this work with that from the literature: CaCO₃ pretreatment produced approximately 750 NL CH₄/kg of methane. Finally, it is important to note that even simple dilution of olive oil waste products with tap water can increase biogas production by 57-fold, which could be a valuable starting option for improving its biodegradability. As the efficiency values were lower than 25% for the entire pretreatment panel, it is necessary to develop additional approaches that can efficiently harvest additional energy from these substrates. Two such options may be either the design of a two-step AD producing both hydrogen and methane or the use of bioreactor configurations other than the Stirred Tank Reactor.

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References

- [1] F. Battista, B. Ruggeri, D. Fino, F. Erriqueus, L. Rutigliano, D. Mescia, Toward the scale-up of agro-food feed mixture for biogas production, *J. Environ. Chem. Eng.* 1 (2013) 1223–1230.
- [2] A. Aveni, Biogas recovery from olive-oil mill waste water by anaerobic digestion, in: *Anaerobic Digestion and Carbohydrate Hydrolysis of Waste*, Elsevier Applied Science Publishers, 1984, pp. 489–491.
- [3] B. Rincon, E. Sanchez, F. Raposo, R. Borja, L. Travieso, M.A. Martin, A. Martin, Effect of the organic loading rate on the performance of anaerobic acidogenic fermentation of two phase olive mill solid residue, *Waste Manage.* 28 (2008) 870–877.
- [4] N. Azbar, A. Bayram, A. Filibeli, A. Muezzinoglu, F. Sengul, A. Ozer, A review of waste management options in olive oil production, *Crit. Rev. Environ. Sci. Technol.* 34 (2004) 209–247.

- [5] R. Borja, J. Alba, A. Martín, A. Mancha, Influencia de la velocidad de carga orgánica sobre el proceso de digestión anaerobia de aguas de lavado de aceitunas de almazara en reactores de lecho fluidizado, *Grasas Aceites* 49 (1) (1998) 42–49.
- [6] A. Roig, M.L. Cayuela, M.A. Sanchez Monedero, An overview on olive mill wastes and their valorisation methods, *Waste Manage.* 26 (2006) 960–969.
- [7] A.R. Tekin, A.C. Dalgic, Biogas production from olive pomace, *Resour. Conserv. Recycl.* 30 (2000) 301–313.
- [8] O.N. Agdag, Biodegradation of olive-mill pomace mixed with organic fraction of municipal solid waste, *Biodegradation* 22 (2011) 931–938.
- [9] M. Carlsson, A. Lagerkvist, F. Morgan-Sagastum, The effects of substrate pretreatment on anaerobic digestion systems: a review, *Waste Manage.* 32 (2012) 1634–1650.
- [10] P. Kianmehr, W. Parker, P. Seto, An evaluation of protocols for characterization of ozone impacts on WAS properties and digestibility, *Bioresour. Technol.* 101 (22) (2010) 8565–8572.
- [11] F. Fdz-Polanco, R. Velazquez, S.I. Perez-Elvira, C. Casas, D. Del Barrio, F.J. Cantero, M. Fdz-Polanco, P. Rodriguez, L. Panizo, J. Serrat, P. Rouge, Continuous thermal hydrolysis and energy integration in sludge anaerobic digestion plants, *Water Sci. Technol.* 57 (2008) 1221–1226.
- [12] M. Beccari, F. Bonemazzi, M. Majone, C. Riccardi, Interaction between acidogenesis and methanogenesis in the anaerobic treatment of olive oil mill effluents, *Water Res.* 30 (1) (1996) 183–199.
- [13] L.M. Nieto, G. Hodaifa, S. Rodriguez, J.A. Gimenez, J. Ochando, Degradation of organic matter in olive-oil mill wastewater through homogeneous Fenton-like reaction, *Chem. Eng. J.* 173 (2011) 503–510.
- [14] G. Pannelli, M. Servili, M. Baldioli, G.F. Montedoro, Changes in the phenolic and pectic substances in olive fruit and oil as a function of ripening, cultivar and extraction technology, *Olea* 21 (1991) 64.
- [15] N. Azbar, T. Keskin, A. Yuruyen, Enhancement of biogas production from olive mill effluent (OME) by co-digestion, *Biomass Bioenergy* 32 (2008) 1195–1201.
- [16] H.K. Obied, M.S. Allen, D.R. Bedgood, P.D. Prensler, K. Robords, R. Stockmann, Bioactivity and analysis of biophenols recovered from olive mill waste, *J. Agric. Food Chem.* 53 (2005) 823–837.
- [17] A. Ramos-Cormenzana, B. Juarez-Jimenez, M.P. Garcia-Pareja, Antimicrobial activity of olive mill wastewaters (alpechin) and biotransformed olive oil mill wastewater, *Int. Biodeterior. Biodegrad.* 38 (1996) 283–290.
- [18] H.K. Obied, Y. Song, S. Faley, M. Loughlin, A. Rehman, R. Mailer, T. Masoul, S. Aybolla, Biophenols and antioxidant properties of Australian canola meal, *J. Agric. Food Chem.* 38 (2013) 9176–9184.
- [19] G. Pekin, S. Haskok, S. Sargin, Y. Gezgün, R. Eltem, E. Ikizoglu, N. Azbar, F.V. Sukan, Anaerobic digestion of Aegean olive mill effluents with and without pretreatment, *J. Chem. Technol. Biotechnol.* 85 (2010) 976–982.
- [20] G. Padovani, C. Pintucci, P. Carlozzi, Dephenolization of stored olive-mill wastewater, using four different adsorbing matrices to attain a low-cost feedstock for hydrogen photo-production, *Bioresour. Technol.* 138 (2013) 172–179.
- [21] T. Coskun, E. Debik, N. Demir, Operational cost comparison of several pretreatment techniques for OMW treatment, *Clean – Soil Air Water* 40 (1) (2012) 95–99.
- [22] T. Shigematsu, Y. Tang, K. Kida, Microbial communities related to methane fermentation processes, *Monograph. Seibutsu-kogaku Kaishi* 87 (12) (2009) 570–596.
- [23] APHA/AWWA/WEF, Standards Methods for the Examination of Water and Wastewater, United Book Press Inc., Baltimore, Maryland, 1998.
- [24] B. Ruggeri, M. Bernardi, T. Tommasi, On the pre-treatment of municipal organic waste towards fuel production: a review, *Int. J. Environ. Pollut.* 49 (2012) 226–250.
- [25] J.D. Browne, E. Allen, D. Murphy, Improving hydrolysis of food waste in a leach bed reactor, *Waste Manage.* 33 (2013) 2470–2477.
- [26] L.T. Fan, Y. Lee, D. Beardmore, Mechanism of the enzymatic hydrolysis of cellulose: effects of major structural features of cellulose on enzymatic hydrolysis, *Biotechnol. Bioeng.* 22 (1980) 177–199.
- [27] R. Uma Rani, S. Adish Kumar, S. Kaliappan, Low temperature therm-chemical pretreatment of dairy waste activated sludge for anaerobic digestion process, *Bioresour. Technol.* 103 (2012) 415–424.
- [28] K. Lee, P. Chntrasakdakul, D. Kim, M. Kong, K.Y. Park, Ultrasound pretreatment of filamentous algal biomass for enhanced biogas production, *Waste Manage.* 34 (6) (2014) 1035–1040.
- [29] T. Kamizela, M. Kowalczyk, The influence of sonication parameters on the disintegration effect of activated sludge, in: *Proceedings of the Conference on Environmental Engineering IV*, Lublin, Poland, 3 September 2012, pp. 191–196.
- [30] J.J. Lay, Y.J. Lee, T. Noike, Feasibility of biological hydrogen production from organic fraction of municipal solid waste, *Water Res.* 33 (11) (1999) 2579–2586.
- [31] Systat Software Inc., SigmaPlot ver.10.01, The simplest way to analyse and graph data, San Jose, CA, 2007.
- [32] B. Ruggeri, S. Sanfilippo, T. Tommasi, Sustainability of (H₂ + CH₄) by anaerobic digestion via EROI approach and LCA evaluations, in: A. Singh, D. Pant, S.I. Olsen (Eds.), *Life Cycle Assessment of Renewable Energy Sources*, Green Energy and Technology Series, Springer, New York, 2013, pp. 169–194.
- [33] D.L. Nelson, M. Cox, *Lehninger Principles of Biochemistry*, McMillan, New York, 2008.
- [34] B. Ruggeri, A.C. Luongo Malave, M. Bernardi, D. Fino, Energy efficacy used to score organic refuse pretreatment processes for hydrogen anaerobic production, *Waste Manage.* 33 (2013) 2225–2233.
- [35] Y. Chen, J.J. Cheng, K.S. Creamer, Inhibition of anaerobic digestion process: a review, *Bioresour. Technol.* 99 (2008) 4044–4064.
- [36] D. Deublein, A. Steinhauser, *Biogas From Waste and Renewable Resources*, Wiley-VCH, Verlag GmbH & Co., Weinheim, 2011.
- [37] A. Ginos, T. Manios, D. Mantzavinos, Treatment of olive mill effluents by coagulation–flocculation–hydrogen peroxide oxidation and effect on phytotoxicity, *J. Hazard. Mater.* B133 (2006) 135–142.
- [38] N. Azbar, T. Keskin, E.C. Catalkaya, Improvement in anaerobic degradation of olive mill effluent (OME) by chemical pretreatment using batch systems, *Biochem. Eng. J.* 38 (2007) 379–383.
- [39] C.A. Jackson-Moss, J.R. Duncan, D.R. Cooper, The effect of calcium on anaerobic digestion, *Biotechnol. Lett.* 11 (3) (1989) 219–224.
- [40] L. Liu, J. Meiting, W. Li, Q. Hou, Dissolution of cellulose from AFEX-pretreated *Zoysia japonica* in AMIMCl with ultrasonic vibration, *Carbohydr. Polym.* 98 (2013) 412–420.